

A-level
ECONOMICS
Paper 1 Markets and market failure
Predicted Paper Set B — Hard
Environmental economics focus

MARK SCHEME and MODEL ANSWERS

Total paper mark: 80 | Section A: 40 marks | Section B: 40 marks

General marking guidance

- Examiners reward responses for what candidates have demonstrated, not penalised for omissions.
- Where a response falls between two levels, the mark awarded should reflect the quality of the analysis and evaluation, not just the number of points made.
- Diagrams must be accurate and fully labelled to score full marks. A diagram that is described clearly in words may earn partial credit.
- Indicative content is not exhaustive. Candidates may offer alternative relevant arguments and examples which should be credited on merit.
- For evaluation questions, credit the quality of judgement — a clear, reasoned conclusion supported by analysis is essential for the top level.

Assessment Objectives (AOs)

AO1 Knowledge and understanding of economic terms, concepts, theories and models.

AO2 Application of economic knowledge and understanding to various contexts.

AO3 Analysis of economic issues, showing understanding of their impact on economic agents.

AO4 Evaluation of economic arguments and use of qualitative and quantitative evidence to support informed judgements.

Section A — Data Response

Carbon Trading and UK Industrial Emissions

0 1	Calculate the percentage fall in UK industrial emissions, 2019–2024.	2 marks
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Indicative content

Formula: Percentage change = $((\text{New} - \text{Old}) / \text{Old}) \times 100$

Working: $((79 - 102) / 102) \times 100 = (-23 / 102) \times 100 = -22.549\% \approx -22.5\%$

Answer: A fall of **22.5%** (accept 22.5% to 1 d.p., or -22.5%).

Mark allocation

- **1 mark** — correct numerical answer of 22.5% (accept 22.5% or 22.55% rounded).
- **1 mark** — correctly identifying the change as a fall / decrease (or a negative value correctly signed).

Award 2 marks for a fully correct answer of – 22.5% (a fall) with or without working shown.
Award 1 mark for correct working but wrong final figure, or for the correct figure without the direction stated.

0 2	Diagram and explanation: how a tightening of the cap raises the carbon permit price.	4 marks
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Indicative diagram

A supply-and-demand diagram for tradable pollution permits with:

- Axes: Price of permits (£ per tonne of CO₂) on the vertical axis, Quantity of permits on the horizontal axis.
- Supply curve is perfectly inelastic (vertical) at the cap Q₁, because the regulator fixes the total number of allowances.
- Downward-sloping demand curve (firms' willingness to pay for the right to emit CO₂).
- Initial equilibrium at price P₁, quantity Q₁.
- Supply shifts left to a new vertical line at Q₂ < Q₁ when the cap is tightened.
- New equilibrium at P₂ > P₁, quantity Q₂.

Indicative written explanation

The regulator sets the maximum number of allowances equal to the cap, making supply perfectly inelastic. Firms demand allowances because they need them to cover actual emissions; this demand is downward-sloping because at higher prices firms find it worthwhile to abate rather than buy permits. When the regulator tightens the cap, the vertical supply curve shifts to the left. Holding demand constant, this creates excess demand at the old price and bids the market price up to P₂. The higher permit price makes emissions-

intensive production more costly, encouraging firms to invest in abatement or switch to cleaner inputs.

Mark allocation (4 marks)

- **1 mark** — correctly labelled axes (Price and Quantity of permits).
- **1 mark** — perfectly inelastic (vertical) supply curve at Q_1 with a correctly drawn downward-sloping demand curve.
- **1 mark** — leftward shift of supply to Q_2 with new equilibrium at P_2 .
- **1 mark** — written explanation of the mechanism (tighter cap → reduced supply of permits → excess demand at P_1 → higher permit price P_2).

A clear verbal description of the diagram may earn up to 3 marks where a visual diagram is absent but the mechanism is fully explained.

03	Analyse two ways in which the UK ETS may correct the market failure associated with industrial pollution.	9 marks
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Required economic content

- Definition of market failure — the free market produces too much of a good whose production generates negative externalities; marginal social cost (MSC) exceeds marginal private cost (MPC).
- Industrial emissions are a classic negative externality of production — the firm does not bear the full social cost of the CO₂ it releases (climate damage, air quality, health).
- The ETS internalises this externality by forcing firms to pay for each tonne of CO₂ they emit.

Way 1: Internalising the externality through a carbon price

Each tonne of CO₂ emitted now carries an explicit price (the permit price). This raises firms' marginal private cost so that it approaches marginal social cost. On a standard externality diagram the supply curve shifts upwards by the marginal external cost, reducing equilibrium output from the privately optimal Q_1 towards the socially optimal Q^* . Extract B provides direct evidence: as the permit price rose from £22 (EU ETS, 2019) to £78 in 2024, UK industrial emissions fell from 102 to 79 MtCO₂ — a 22.5% reduction. Government modelling cited in Extract C suggests emissions would have been 9–14 MtCO₂ higher without the ETS, confirming the corrective effect.

Way 2: Dynamic efficiency — incentivising low-carbon investment and innovation

A positive carbon price creates an ongoing profit motive to invest in abatement technologies and switch to lower-emission inputs. Firms whose marginal abatement cost is below the permit price will abate rather than buy permits; firms whose abatement cost is above the price will buy permits and use them to cover continuing emissions. This trade equalises marginal abatement costs across the economy, delivering abatement at least cost. The rising renewable share of electricity generation shown in Extract B (37% in 2019 to 47% in 2024) is consistent with this mechanism: the effective cost of fossil-based generation has been pushed up, making investment in renewables relatively more attractive.

Application using the extracts

- **Extract A:** permit price tripled from £50/tonne (2021) to £78/tonne (mid-2024) — evidence of a meaningful price signal.
- **Extract B:** emissions fell by 22.5% while renewable share rose by 10 percentage points — evidence for both corrective mechanisms.
- **Extract C:** government modelling suggests emissions would have been 9–14 MtCO₂ higher in 2024 without the ETS — quantifies the corrective effect.

Mark scheme — levels

Level	Marks	Descriptor
Level 3	7–9	Two distinct, fully-developed chains of reasoning linking the ETS mechanism to the correction of a negative externality. Accurate diagram support where relevant. Clear and sustained use of the extracts, including data citation.
Level 2	4–6	Two ways identified but at least one is under-developed, or only one is explored in depth. Some use of the extracts. Minor inaccuracies in economic analysis.
Level 1	1–3	One mechanism identified; limited development. Diagram missing or inaccurate. Extracts referenced descriptively or not at all.

Maximum 6 marks if only one mechanism is explored in depth, even if excellent. Reward any economically valid second mechanism, such as revenue generation (hypothecation to green investment), providing certainty over total emissions, or driving reformulation of supply chains.

04

Evaluate the view that tradable pollution permits are a more effective policy than indirect taxation for reducing industrial carbon emissions.

**25
marks**

Arguments supporting tradable permits (FOR)

- **Quantity certainty:** the regulator fixes the cap, guaranteeing a maximum level of emissions. A carbon tax, by contrast, fixes price but leaves total emissions uncertain. For a legally binding Net Zero target, quantity certainty is directly aligned with the policy objective.
- **Least-cost abatement:** trading equalises marginal abatement costs across firms. Those with cheap abatement options cut first and sell surplus permits to firms with high abatement costs. This is allocatively efficient and minimises the total cost of hitting the cap.
- **Empirical evidence:** Extract B shows UK industrial emissions fell by 22.5% between 2019 and 2024 while the renewable share of generation rose from 37% to 47%. Extract C cites government modelling suggesting 9–14 MtCO₂ of additional emissions were avoided because of the scheme.
- **Dynamic efficiency:** the prospect of selling surplus allowances gives firms a profit motive to innovate beyond the required level of abatement — stronger than a simple tax liability.
- **Revenue with tightening capacity:** government auctions of allowances generate hypothecatable revenue, and the cap can be tightened over time to deliver declining emissions.

Arguments against permits / in favour of carbon taxation (AGAINST)

- **Price volatility and investment uncertainty:** Extract C reports that 41% of energy-intensive firms had postponed planned low-carbon investment because of allowance price uncertainty. A carbon tax provides a predictable price signal, supporting long-term capital planning.
- **Over-allocation risk:** Extract A notes environmental groups argue the UK cap is too generous. The early EU ETS suffered from exactly this, producing low prices and weak incentives. Taxes avoid this political-economy risk.
- **Administrative complexity:** permit markets require extensive monitoring, reporting and verification, plus a functioning auction and secondary market — costly and difficult for smaller emitters. A tax can be applied at point of fuel supply with minimal new infrastructure.
- **Coverage:** the UK ETS covers only large installations (electricity, heavy industry, aviation). Small and dispersed emitters (road transport, heating) are more straightforwardly taxed through fuel duty.
- **Carbon leakage:** Extract A warns that high carbon prices may push production to jurisdictions with weaker regulation — reducing UK emissions without cutting global emissions. Both policies share this risk, but a domestic tax combined with a carbon border adjustment can mitigate it.

Relevant extensions and alternatives

- **Hybrid instruments:** the UK ETS operates with a minimum auction reserve price, providing a floor that functions like a partial tax and reduces downside volatility.

- Carbon Border Adjustment Mechanisms (UK CBAM from 2027) address leakage for both instruments.
- Complementary regulation is often required for non-price-sensitive decisions — e.g. the UK's 2035 ban on new petrol/diesel cars, or building regulations on insulation standards.
- Elasticity matters: where demand for emissions-intensive output is price-inelastic in the short run, neither instrument produces rapid abatement — long-run capital turnover drives the reduction.

Expected judgement and conclusion

Top-band answers should recognise that permits and taxes are two sides of the same coin: both attach a price to CO₂ and operate through the MPC-to-MSC margin. The difference is which variable is fixed — quantity or price. A reasoned judgement should note that permits suit legally binding emission targets (such as UK Net Zero) where quantity certainty matters most, while taxes offer price certainty which is valuable when the overriding concern is investment signalling. The UK case study suggests a well-designed ETS with a price floor has performed well, but the CBI evidence on investment postponement suggests that price volatility remains a genuine cost. The strongest conclusion is therefore that neither is universally superior: the best policy for reducing industrial carbon emissions is probably a price-floored ETS for large installations, complemented by fuel-based carbon taxation for dispersed emitters and targeted regulation where price signals fail.

Mark scheme — levels

Level	Marks	Descriptor
Level 5	21–25	Sophisticated analysis and evaluation. Sustained chains of reasoning; accurate diagram(s); substantial use of both extracts and own knowledge; clear, well-supported judgement that addresses the 'more effective' comparison directly.
Level 4	16–20	Good analysis and evaluation. Diagram present and mostly accurate. Two or more developed evaluative points on each side. Supported judgement, possibly less nuanced.
Level 3	11–15	Sound analysis with some evaluation. Use of extracts is present but may be descriptive. Comparison between permits and taxes may be uneven or judgement asserted.
Level 2	6–10	Some relevant analysis. Limited evaluation. Weak or partial use of extracts; comparison underdeveloped.
Level 1	1–5	Limited analysis; evaluation absent or generic. Little or no use of extracts.

Section B — Essay

Environmental externalities and policy design

05	<i>Explain, using a diagram, how negative externalities of production lead to allocative inefficiency.</i>	15 marks
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Required definitions

- **Negative externality of production:** a spillover cost imposed on a third party that is not reflected in the private costs of the producer. The marginal social cost (MSC) therefore lies above the marginal private cost (MPC).
- **Allocative efficiency:** occurs when resources are allocated such that marginal social benefit (MSB) equals marginal social cost (MSC). At this point no reallocation could make society better off.
- **Market failure:** the free market fails to allocate resources at the socially optimal level, producing a welfare loss.

Required diagram

A standard negative-production-externality diagram showing:

- Axes: Price/Cost on the vertical axis, Quantity on the horizontal axis.
- MPC curve (upward-sloping) — the private supply curve.
- MSC curve drawn above MPC by the amount of the marginal external cost (MEC), diverging or parallel.
- Demand = MPB = MSB (assuming no consumption externality), downward-sloping.
- Free-market equilibrium at Q_1 where $MPC = D$, with price P_1 .
- Social optimum at $Q^* < Q_1$ where $MSC = D$.
- Welfare loss (deadweight loss) triangle shaded between Q^* and Q_1 , bounded above by MSC and below by MPB.

Required written explanation

At the free-market equilibrium Q_1 , producers equate their private marginal cost with market price; they ignore the external cost imposed on third parties (for example CO₂ emissions harming the climate). Because the true cost of the last unit produced — the marginal social cost — exceeds the marginal social benefit at Q_1 , society would be better off if output were lower. Specifically, every unit between Q^* and Q_1 adds more to total social cost than to total social benefit, and the cumulative loss is measured by the welfare-loss triangle between MSC and MSB over the range Q^* to Q_1 .

Allocative inefficiency therefore arises because resources are being devoted to the production of units whose social cost exceeds social benefit. The price mechanism fails to transmit the full cost of production, leading to overproduction of the polluting good and underproduction of the unpolluted alternative. Real-world examples include industrial CO₂ emissions, noise and air pollution from road haulage, and effluent discharge from manufacturing into rivers.

Expected content for top-band responses

- Accurate definitions of MPC, MSC, MEC, MPB/MSB.
- Fully labelled diagram with welfare loss triangle clearly identified.
- Clear chain of reasoning from the divergence of MPC and MSC to overproduction at Q_1 , and the resulting welfare loss.
- At least one applied example (e.g. industrial emissions, power station pollution, factory effluent).

Mark scheme — levels

Level	Marks	Descriptor
Level 4	13–15	Fully developed explanation with accurate, well-labelled diagram and clear welfare-loss triangle. Sustained chain of reasoning from the externality to allocative inefficiency. Appropriate real-world example. Terminology precise throughout.
Level 3	9–12	Sound explanation with a mostly accurate diagram. Chain of reasoning is clear but may have a minor gap or imprecision. Some real-world application.
Level 2	5–8	Some understanding. Diagram present but partially inaccurate or incompletely labelled. Allocative inefficiency asserted rather than explained.
Level 1	1–4	Descriptive only. Diagram missing or significantly flawed. Little use of MSC/MPC framework.

06

Evaluate: “Market-based environmental policies are always more efficient than command-and-control regulation.”

25
marks

Key definitions

- **Market-based policies:** instruments that attach a price to pollution, leaving firms to choose their response — e.g. carbon taxes, tradable pollution permits, subsidies for clean technology.
- **Command-and-control regulation:** direct legal requirements specifying either the technology firms must use or the quantity of pollution permitted — e.g. emissions standards, technology mandates, outright bans.

Arguments in favour of the statement (*FOR* market-based)

- **Allocative efficiency:** market instruments equalise marginal abatement cost across firms. Those with the cheapest abatement options act first; high-cost firms pay. Command-and-control typically imposes a uniform standard regardless of cost differences, leading to unnecessarily high total abatement costs.
- **Dynamic efficiency:** a carbon price or tradable permit creates an ongoing incentive to innovate because reducing emissions beyond the standard is financially rewarded. A technology mandate, by contrast, removes the incentive to go further once the standard is met.
- **Flexibility:** firms can choose their own method of compliance — fuel-switching, efficiency gains, abatement technology, or output reduction — reducing the information burden on government.
- **UK examples:** UK ETS (industrial emissions fell 22.5% from 2019–2024); Carbon Price Support added to ETS for power-sector; SDIL sugar levy; plastic bag charge reduced single-use bag use by over 95%.

Arguments against the statement (*AGAINST* market-based / *FOR* regulation)

- **Price volatility:** fluctuating carbon or permit prices can undermine long-term investment. The 2024 CBI survey found 41% of energy-intensive UK firms postponed low-carbon investment because of allowance price uncertainty — regulation provides a clearer planning horizon.
- **Hazardous and local pollutants:** for pollutants with concentrated local effects (e.g. NO_x near schools, particulates in cities), uniform pricing may leave dangerous local concentrations. Direct limits are preferable — hence the London ULEZ and its 2023 outer expansion.
- **Regressive incidence:** carbon pricing raises energy bills, disproportionately affecting low-income households. Regulation that shifts costs onto firms or vehicle manufacturers can be less visibly regressive.
- **Administrative complexity:** functioning permit markets require monitoring, auctions, and secondary trading — costly for small emitters. A blanket regulatory ban (e.g. incandescent light bulbs phased out in 2012; new petrol/diesel cars banned from 2035) can be simpler and quicker.
- **Political economy risks:** market-based schemes can be watered down via over-generous caps (as Extract A suggests some argue for the UK ETS) or successive tax cuts. Regulation embedded in law can be more durable.

- **Coverage gaps:** UK ETS covers only large installations; regulation (building energy-efficiency standards, Euro 6 emission limits) reaches sectors a carbon price does not easily touch.

Relevant UK examples to deploy

- UK ETS (market-based) — covered in Section A data.
- 2035 ban on the sale of new petrol and diesel cars (regulation).
- London ULEZ and its 2023 outer-London expansion (regulation on vehicle emissions).
- Plastic bag charge since 2015 (market-based nudge) — reduced bag use by more than 95%.
- Building regulations (Part L) on insulation and energy efficiency (regulation).
- Clean Air Act precedents (regulation) that effectively ended London smog.
- Boiler Upgrade Scheme grants for heat pumps (subsidy — market-based).

Expected judgement and conclusion

A strong answer should argue that ‘always’ makes the statement too strong. Market-based policies excel in settings with many diverse emitters, measurable uniform pollutants (CO₂), and mature monitoring infrastructure — hence the UK ETS delivers lower abatement cost than a one-size-fits-all standard would. However, regulation remains more efficient when pollutants have concentrated local effects, when decision-makers are insensitive to price (habitual behaviour, split incentives in rental housing), when the pollutant is too dangerous to allow any trade, or when the policy goal is a hard sectoral transition such as ending ICE car sales. In practice the most successful UK environmental policy stack is hybrid: a price-floored ETS for industry and power, fuel duties for transport, and regulatory mandates for technology transitions and building standards. The conclusion should therefore reject ‘always’: each instrument has its domain, and combining them exploits their comparative strengths.

Mark scheme — levels

Level	Marks	Descriptor
Level 5	21–25	Sophisticated, balanced analysis and evaluation. Uses multiple UK examples (at least two on each side) with accurate economic reasoning. Clear, nuanced judgement addressing the word ‘always’ explicitly. Diagrams where helpful.
Level 4	16–20	Good analysis and evaluation. Uses at least two UK examples. Supported judgement, possibly less developed.
Level 3	11–15	Sound analysis with some evaluation. One or two UK examples. Judgement may be asserted rather than reasoned.
Level 2	6–10	Limited analysis. Evaluation weak or superficial. Few examples.
Level 1	1–5	Descriptive only. Little or no evaluation.

Summary of Marks

Question	Topic	AO focus	Marks
0 1	Quantitative skill — percentage change	AO2	2
0 2	Tradable permits — supply and demand diagram	AO1, AO2	4
0 3	UK ETS correcting market failure	AO1, AO2, AO3	9
0 4	Permits vs carbon taxation — evaluation	AO1–AO4	25
0 5	Negative externalities of production	AO1, AO3	15
0 6	Market-based vs regulatory environmental policy	AO1–AO4	25
TOTAL	Paper 1: Markets and Market Failure	—	80

END OF MARK SCHEME